

realisation of a double-layer perceptron, an element that can perform classification functions after an appropriate learning procedure. The main difficulty the research team faced was in understanding the complex electrochemical interplay that is the basis for memristive behaviour, which would give them the means to control it.

Researchers addressed this challenge by using commercial polymers and modifying their electrochemical properties at the macroscopic level. The most surprising result was that it was possible to check the electrochemical response of the device by changing the formulation of gels acting as polyelectrolytes, allowing for the study of ionic exchanges related to the biological object, which activates the electrochemical response of the conductive polymer. These developments open the way to make compatible polyaniline based devices with an interface that should be naturally-, biologically- and electrochemically-compatible and functional.

Pectin is a key ingredient for making delicious jellies and jams, not as a component for a complex hybrid device that links biological and electronic systems. But a team of scientists has built on previous work in this field using pectin with a high degree of methylation as the medium to create a new architecture of a hybrid device with a double-layered polyelectrolyte that alone drives memristive behaviour.

The team of researchers has explained the creation of the hybrid device. In this research, they applied materials generally used in the pharmaceutical and food industries in electrochemical devices. The idea of using the buffering capability of these bio-compatible materials as solid polyelectrolyte is completely innovative, and it is the first time that these bio-polymers have been used in devices based on

organic polymers and in a memristive device.

The next steps are interfacing the memristor network with other living beings, for example, plants and, ultimately, the realisation of hybrid systems that can learn and perform logic/classification functions.

Using phase-change material.

A nano-sized device made from a chalcogenide performs neuron-like calculations. Electrical pulses convert a doped chalcogenide from an amorphous phase into a crystalline one. Once a certain amount has been converted, conductance of the material suddenly jumps, mimicking the firing of a neuron.

A research team has reported that nano-sized devices made from phase-change materials can mimic how neurons fire to perform certain calculations. This report shows quite concretely that we can make simple but effective hardware mimics of neurons, which could be made really small and therefore have low operating powers. The device imitates how an individual neuron integrates incoming signals from other neurons to determine when it should fire. These input signals change the electrical potential across the neuron's membrane—some increase it, others decrease it. Once that potential passes a certain threshold, the neuron fires.

Previously, engineers have mimicked this process using combinations of capacitors and silicon transistors, which can be complex and difficult to scale down. The new work demonstrates a potentially simpler system that uses a phase-change material to play the part of a neuron's membrane potential.

The doped chalcogenide $\text{Ge}_2\text{Sb}_2\text{Te}_5$, which has been tested in conventional memory devices, can exist in two phases: glassy amorphous and crystalline. Electrical pulses slowly convert the material from amorphous to crystalline,

which, in turn, changes its conductance. At a certain level of phase change, the material's conductance suddenly jumps, and the device fires like a neuron.

Last words

Different material systems or devices have strengths in various characteristics and required characteristics for a synaptic device that will enable the development of brain-inspired systems technology that scales to biological levels of device density, parallelism and functionality. The synaptic device is a simple two-terminal nano-scale device that can reach brain-level parallelism and compactness.

Main characteristics are the ones that are directly related to parallelism, energy efficiency and fault tolerance. The synaptic device should emulate plasticity by implementing an analogue-like transition between different conductance states with very low energy consumption per synaptic event. Exact performance metrics depend on the target application and the scale of the system design.

Although most of the work focuses on some specific applications for synaptic devices, the field has the potential to be utilised in a broad range of computing applications, especially the ones at the intersection of sensing and computation, where real-time and parallel processing of large-scale data is crucial.

There is an enormous opportunity to completely rethink the design of computational systems in order to gain orders of magnitude of improvement in computational efficiency through inspiration from the biological brain. More interactions among different research disciplines—devices, circuits, architecture and computing—can further cultivate the synaptic electronics field and help define more targeted research paths for the future. **EFY**